A. Ring road

time limit per test: 2 seconds
memory limit per test: 256 megabytes

Nowadays the one-way traffic is introduced all over the world in order to improve driving safety and reduce traffic jams. The government of Berland decided to keep up with new trends. Formerly all n cities of Berland were connected by n two-way roads in the ring, i. e. each city was connected directly to exactly two other cities, and from each city it was possible to get to any other city. Government of Berland introduced one-way traffic on all n roads, but it soon became clear that it's impossible to get from some of the cities to some others. Now for each road is known in which direction the traffic is directed at it, and the cost of redirecting the traffic. What is the smallest amount of money the government should spend on the redirecting of roads so that from every city you can get to any other?

Input

The first line contains integer n $(3 \le n \le 100)$ — amount of cities (and roads) in Berland. Next n lines contain description of roads. Each road is described by three integers a_i, b_i, c_i $(1 \le a_i, b_i \le n, a_i \ne b_i, 1 \le c_i \le 100)$ — road is directed from city a_i to city b_i, redirecting the traffic costs c_i.

Output

Output single integer — the smallest amount of money the government should spend on the redirecting of roads so that from every city you can get to any other.

Examples

input Copy

```
3
1 3 1
1 2 1
3 2 1
```

output Copy

```
1
```

input Copy

```
3
1 3 1
1 2 5
3 2 1
```

output Copy

```
2
```

input Copy

```
6
1 5 4
5 3 8
2 4 15
1 6 16
2 3 23
4 6 42
```

output Copy

```
39
```

input Copy

```
4
1 2 9
2 3 8
```

→ **Attention**

The package for this problem was not updated by the problem writer or Codeforces administration after we've upgraded the judging servers. To adjust the time limit constraint, a solution execution time will be multiplied by 2. For example, if your solution works for 400 ms on judging servers, then the value 800 ms will be displayed and used to determine the verdict.

Codeforces Beta Round 24

Finished

Practice

→ **Virtual participation**

Virtual contest is a way to take part in past contest, as close as possible to participation on time. It is supported only ICPC mode for virtual contests. If you've seen these problems, a virtual contest is not for you - solve these problems in the archive. If you just want to solve some problem from a contest, a virtual contest is not for you - solve this problem in the archive. Never use someone else's code, read the tutorials or communicate with other person during a virtual contest.

Start virtual contest

→ **Clone Contest to Mashup**

You can clone this contest to a mashup.

Clone Contest

→ **Submit?**

Language: GNU G++23 14.2 (64 bit, msys2

Choose file: Choose File No file chosen

Submit

→ **Last submissions**

Submission	Time	Verdict
372125028	Apr/22/2026 14:14	Accepted
372124482	Apr/22/2026 14:09	Accepted
372124264	Apr/22/2026 14:07	Accepted
372123962	Apr/22/2026 14:03	Wrong answer on test 1

```
3 4 7
4 1 5
```

output Copy

```
0
```

372123833	Apr/22/2026 14:02	Wrong answer on test 1
372123572	Apr/22/2026 14:00	Wrong answer on test 1
372123413	Apr/22/2026 13:58	Wrong answer on test 3

→ Problem tags

graphs *1400

No tag edit access

→ Contest materials

- Announcement
- Tutorial

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